

A Systematic Review of the Cost and Cost-Effectiveness Studies of Proton Radiotherapy

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Key Words: proton radiation therapy, cost-effectiveness, healthcare economics

Shortened Title: Cost-Effectiveness of Protons

Acknowledgements: We thank Dr. David Sher for critical analysis and review of the manuscript.

Funding: None

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Precis: In the face of economically changing landscapes of healthcare reform, having evidence-based justification of new technologies, such as proton radiation therapy, is essential. We show that this modality is not the most cost-effective option for prostate or early-stage lung cancers; however, it is the most cost-effective for several pediatric cancers and remains favorable for some left-sided breast cancers, loco-regionally advanced lung cancers, and select head/neck cancers.

This is the author manuscript accepted for publication and has undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the [Version record](#). Please cite this article as [doi:10.1002/cncr.29882](https://doi.org/10.1002/cncr.29882).

CONFLICT OF INTEREST STATEMENT

MPM has served as a consultant for Abbott, Bristol-Meyers-Squibb, Celldex, Cavion, Elekta, Novartis, Novocure, and Roche, has research funding from Novocure and Cellectar, and has served in a leadership capacity on the Pharmacyclics BOD (with stock options).

MVM served on the PCORI Patient Engagement Advisory Panel and also has grant support from the American Society for Radiation Oncology (ASTRO).

Accepted Article

ABSTRACT

PURPOSE: Economic analyses of new technologies such as proton beam radiotherapy (PBT) are a public health priority. No systematic review of PBT's cost-effectiveness (CE) has been performed.

METHODS: Systematic searches of PubMed, EMBASE, abstracts from American Society for Radiation Oncology and American Society of Clinical Oncology meetings, and the Cost-Effectiveness Analysis Registry was conducted (2000-2015); along with the Particle Therapy Co-Operative Group of North America for both years of existence (2014-2015). Eighteen original investigations were analyzed.

RESULTS: CE for prostate cancer, the most common diagnosis currently treated with PBT, is suboptimal. PBT is the most cost-effective option for several pediatric brain tumors. PBT costs for breast cancer are increased, but favorable for appropriately-selected left-sided cancers at high-risk of cardiac toxicity, and versus brachytherapy for accelerated partial breast irradiation. For non-small cell lung cancer (NSCLC), greatest CE benefits using PBT are observed for loco-regionally advanced –but not early-stage– tumors. PBT offers superior CE in selected head/neck cancer patients at higher risk of acute mucosal toxicities. Similar CE is seen for PBT, enucleation, and plaque brachytherapy for uveal melanoma.

CONCLUSIONS: With limited data, PBT offers promising CE for pediatric brain tumors, well-selected breast cancers, locoregionally advanced NSCLC, and high-risk head/neck cancers. Heretofore, PBT has not been shown cost-effective for prostate cancer or early-stage NSCLC. Careful patient selection is absolutely critical to assess CE. Together with increasing PBT availability, clinical trial evidence, and ongoing major technological improvements, CE data and conclusions from this analysis could change rapidly.

Accepted Article

INTRODUCTION

Clinical application of proton beam radiotherapy (PBT) to treat various cancers is growing rapidly. Ten years ago, there were only four operational PBT facilities in the U.S.; as of the writing of this manuscript, 15 U.S. centers had become operational, with an estimated 91 operational facilities worldwide by 2020.¹ Dosimetric benefits of PBT are well-established – the absence of exit dose beyond the Bragg peak spares normal tissue that would otherwise receive exit dose if a photon beam were employed²; this reduces the whole-body integral dose. Although such data provide convincing rationale for PBT utilization, clinical significance of these findings has not been clearly demonstrated. Currently, PBT is increasingly being used for the treatment of pediatric³, skull base⁴⁻⁵, hepatocellular⁶, head/neck tumors⁷, central nervous system tumors, selected breast and lung cancers, prostate cancer (albeit controversial)⁸, pancreatic carcinoma, and ocular tumors.⁹

Due to rising costs of cancer care, there is concern that conventional PBT centers may not be sustainable in the future.¹⁰⁻¹¹ Others have argued that due to incomplete clinical and toxicity data for PBT versus photons, PBT's CE will be difficult to truly assess.¹²⁻¹³ Furthermore, because PBT technology and delivery is quickly being better optimized, presumed cost-variables themselves have become inaccurate, rendering prior cost comparisons obsolete.¹⁴⁻¹⁵

Studying CE of PBT is difficult, in part because there is great reliance on clinical outcomes and toxicity data, of which there is a clear dearth in PBT. CE studies commonly consist of two types of analyses: modeling studies and non-modeling studies. Modeling studies involve probabilistic input of various future occurrences (e.g. outcomes & toxicities) in the natural history of a group of hypothetical newly-diagnosed cancer patients. These probabilities are not utilized in non-modeling studies (thus being only “cost” analyses), and so though non-modeling studies are technically easier to perform, reliability of data is relatively less.

Two common techniques used to execute modeling analyses involve population-based (Markov) or individual-based (Monte Carlo) modeling and simulation. In either technique, various parameters of the analyzed individuals are specified (e.g., tumor stage, age, RT dose, systemic therapy). Costs of intervention are typically calculated from insurance reimbursement figures and are often used as comparator arms (e.g. protons versus photons). After the intervention takes place in the simulated population, probabilities of adverse effects (and costs of supportive management thereof) with each RT modality, along with corresponding medical costs (e.g. from literature estimates) come into effect as time progresses through successive iterations. Similarly, costs associated with acute/late toxicities, treatment complications, length of hospital stay, and disease course (all based on literature estimates) come into play probabilistically as the lifetime of the simulated individuals progresses until a finite time point or death (from cancer or any cause). Thereafter, RT modalities are compared for overall costs, which takes into account not only treatment costs but also of all aftereffects. A major outcome of these studies includes quality and quantity of further survival, termed quality-adjusted life-years (QALYs), as well as the costs associated with each QALY gained from each RT modality, termed the incremental cost-effectiveness ratio (ICER). The ICER is a substitute for whether a particular intervention is labeled 'cost-effective' by various parties and has been historically estimated to be \$50,000/QALY, also known as the "willingness to pay" (WTP) threshold. Sensitivity analyses are performed thereafter to measure variability in computed costs associated with changes in input parameters.

In this truly contentious and rapidly-evolving environment, evidence-based justification for using new treatment modalities is essential. We provide the first known comprehensive review analyzing all available data regarding PBT cost-effectiveness.

MATERIALS AND METHODS

This systematic review was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹⁶ Eligibility criteria included published work in English evaluating the cost-effectiveness of proton radiation therapy. Sources of information included PubMed, EMBASE, abstracts from annual meetings of the American Society for Radiation Oncology (ASTRO) and the American Society of Clinical Oncology (ASCO), the Cost-Effectiveness Analysis Registry (<https://research.tufts-nemc.org/cear4/>), abstracts from the Particle Therapy Co-Operative Group of North America (PTCOG-NA) for both years of existence (2014-2015), those found in references from the major articles identified, and articles known to the authors. Searches were conducted to identify all articles addressing cost-effectiveness and economics of proton radiotherapy with the following headings: protons, proton facility, proton radiation, cost, cost-effectiveness, value, economics, policy, monetary, reimbursement, and medical insurance. Search terms were restricted from 2000-2015 in efforts to include more recent data; all searches were completed by June 1, 2015. Based on initial searches, 563 articles/abstracts were identified (Figure 1). Articles were reviewed by all authors (VV, MVM, MPM); resolution of discrepancies favored adding the particular manuscript into the analysis, and there was no standard data abstraction form. Care was taken to ensure that inclusion criteria were sufficiently broad such that possibly pertinent publications were excluded by individual screening rather than the initial database search. If abstracts and corresponding journal publications were from the same group, the abstract was excluded. After duplicates were removed, each of the 555 remaining eligible items were independently screened for the criteria described below, and 497 were further excluded. Specifically, articles without specific assessments/reflections on costs of proton radiotherapy, thus being outside the scope of this review, were excluded. Additionally, editorials/commentaries were excluded. Of the fifty-eight publications remaining, 30 were review articles; though many were cited, they were not included in the primary analysis. An additional seven original papers and three abstracts prominently addressed cost-effectiveness of PBT but did not directly assess monetary aspects; these are

cited but not discussed in depth. Thus, eighteen original investigations (3 abstracts and 15 manuscripts) were found to have sufficient focus and relevance to be incorporated.

RESULTS

Prostate Cancer

Prostate cancer is currently the most common cancer in the USA for which PBT is utilized. Peeters et al.¹⁷ found PBT slightly cheaper than prostate IMRT in a nonmodeling study of prostate cancer patients, €16,090 (\$19,469) to €18,160 (\$21,974). Prior to an early Swedish study¹⁸ that showed an increase of €7,952 and 0.297 QALYs gained for PBT, prostate cancer was initially studied from a CE perspective in 2007.¹⁹ Intermediate risk disease was modeled using Markov modeling. Because PBT's ability to spare dose to organs-at-risk results allows for dose-escalation to the tumor, the authors assumed that prescribed prostate dose could be safely escalated with PBT from 81 to 91 Gy while still keeping organ-at-risk dose under tolerance limits, which is questionable in light of dose-outcome data from weaker single-institutional data. Despite this assumption as well as lack of toxicity accountability in the Markov model, PBT was not found as cost-effective as IMRT. In a 70-year-old patient, mean costs were judged as \$63,511 (PBT, €56,734) and \$36,808 (IMRT, €32,880), with QALYs of 8.54 and 8.12 years, respectively. In a 60-year-old man, the costs were \$64,989 (€58,054) and \$39,355 (€35,156) with QALYs of 9.91 and 9.45 years, respectively. Subset analysis, albeit with very few PBT patients, of elective PBT in low-risk cancer using SEER data and 2010 Medicare figures²⁰ showed average costs of \$42,772 (€38,208), as compared with \$29,616 (€26,456) for IMRT, around \$14,000 (€12,506) for prostatectomy, \$16,883 (€15,082) for brachytherapy, and \$2,766 (€2,471) for watchful waiting.

Modeling analysis to compare IMRT, SBRT, and PBT was conducted using Markov methodology in 65-year-old men either eligible for radiotherapy or those declining surgery, which presumably incorporated all three major risk groups to some degree.²¹ There were salient methodological flaws including lack of probabilistic multivariate sensitivity analysis and assumption of one-time cost for toxicities without reassessment thereafter. Using RTOG toxicity criteria and systematic searches of acute/late toxicity data, payer's perspective costs were \$24,873 (€22,219) and 8.11 QALYs for SBRT, \$33,068 (€29,540) and 8.05 QALYs for IMRT, and \$69,412 (€62,005) with 8.06 QALYs for PBT. However, these data fail to take into account that PBT can be used to deliver SBRT, another technique by which PBT costs could decrease.

Using 2008-2009 Medicare reimbursements, one report examined 27,647 patients that underwent prostate radiotherapy; 553 patients received PBT and the remainder conventionally-fractionated IMRT.²² At median 12-month follow-up, there were no differences in genitourinary (17.5% IMRT, 18.8% PBT), gastrointestinal (10.2% vs. 9.9%), or other (5.6% vs. 4.5%) toxicities. Median Medicare reimbursement was \$32,428 (€28,968) for PBT and \$18,575 (€16,593) for IMRT; the authors concluded that PBT was not sufficiently cost-effective at similar toxicity levels experienced by both groups. This is important in light of other Markov data²³ that estimate a need to decrease composite toxicity by 41% in order for PBT to be comparably cost-effective with IMRT. These data are depicted in Table 1, and the lack of a proven CE benefit in prostate cancer is important to consider with the relative ubiquity of prostate PBT treatments being performed throughout the USA and world.

Breast Cancer

A Swedish report²⁴, using Markov modeling, analyzed two groups of left-sided breast cancer patients receiving PBT or conventional whole-breast irradiation (WBI). Total costs were €11,248

(\$13,610) and €5,001 (\$6,051), respectively, with similar QALYs of 12.35 and 12.25. This was consistent with another Markov analysis by the same researchers¹⁸ that showed an increase in €5,920 (\$7,163) and 0.17 QALYs gained for PBT. However, when examining a specific population estimated to have twice the risk of non-radiotherapy-related cardiac disease (owing to baseline hypertension, obesity, hypercholesterolemia, cardiac disease history, etc.), the average cost for PBT per QALY gained was nearly halved from €66,608 (\$80,596) to €34,290 (\$41,491), primarily because of reductions in anticipated cardiac morbidity. However, applicability of this study is uncertain in light of modern techniques utilized for cardiac dose sparing in photon-based RT (e.g. dose modulation, active breathing control, deep inspiratory breath hold, and respiratory gating). Additionally, the assumption of 14% severe RT pneumonitis risk is too high for the modern era. Similar analyses with more modern comparisons therefore would be of great interest.

Together with dosimetric data for PBT in accelerated partial breast irradiation (APBI), one report²⁵ used 2003 Medicare estimates to estimate costs for WBI with electron boost (50 Gy/25 fractions + 10 Gy/5 fractions), PBT, and mixed photon-electron (3-field arrangement) treatment. Though the latter cost \$5,300 (€4,734), PBT was only modestly more expensive (\$13,200, €11,792) than WBI (\$10,600, €9,469). However, this study did not account for costs secondary to potential cardiopulmonary side effects and hence is not strictly a “cost-effectiveness” analysis but rather only a “cost” analysis, which is solely applicable to USA Medicare patients.

A more recent CE analysis for APBI, albeit without methodological details available, was presented at PTCOG-NA 2015²⁶ and used 2014 Medicare allowable charges. Whereas IMRT was the most expensive and linac-based three-dimensional conformal radiotherapy (3DCRT) APBI the least (\$6,771, €6,049), Medicare allowable charges for PBT APBI were \$13,883 (€12,402), only marginally greater than conventional WBI (\$13,149, €11,746). APBI using PBT was also less expensive than strut-adjusted volume implant applicators (\$14,859, €13,265) and only slightly more than MammoSite APBI

(\$12,245, €10,938). Hence, current research examining APBI versus WBI could have implications for potential incorporation of PBT. These data are summarized in Table 2.

Non-Small Cell Lung Cancer

Grutters et al.²⁷ analyzed inoperable stage I NSCLC treated with PBT, carbon ion RT, 3DCRT, and stereotactic body radiotherapy (SBRT). Costs were as follows in respective order: €27,567 (\$33,356), €19,215 (\$23,250), €22,696 (\$27,462), and €13,871 (\$16,784). Corresponding QALYs were 2.33, 2.67, 1.98, and 2.59, respectively. Thus, the costs per QALY gained were lowest for SBRT and carbon ions, although the latter cannot be adequately modeled/compared (true U.S. costs for this modality are unavailable). A critique of this report includes large assumptions of similar utility figures for acute and chronic aftereffects as well as lack of assessment beyond 5 years, both owing to the dearth of available clinical data. Further, this report did not segregate patients into central versus peripheral locations, and therefore could not identify possible subsets that could benefit. The data, however, are consistent with a non-modeling paper (with corresponding limitations) published by the same group¹⁷ that calculated literature-based (2003-2005) costs. SBRT was found to be cheapest at €3,720 (\$4,501), as compared with €8,150 (\$9,862) for 3DCRT and €16,090 (\$19,469) for PBT.

Markov analysis of concurrent chemoradiotherapy for locally-advanced NSCLC using PBT, IMRT, and 3DCRT demonstrated that PBT increases QALYs by 0.549 and 0.452 compared to 3DCRT and IMRT, respectively.²⁸ Although such data is hypothesis-generating, future studies are required to determine the CE of PBT for advanced-stage lung cancers. These data are depicted in Table 3, and further reinforced by additional recent clinical data which need to be considered in parallel.

Head/Neck Cancers

Markov modeling in an aforementioned study¹⁸ for head/neck cancers illustrated the cost of PBT to be just €3,887 (\$4,703) higher, with a relatively large 1.02 QALYs gained from PBT. This indicates that the CE of PBT for head and neck cancer could be substantial, although the lack of toxicity data for PBT in head/neck cancers at the time of publication renders questionable applicability. A non-modeling paper¹⁷ calculated literature-based (2003-2005) costs for head/neck cancers. IMRT cost was estimated at €11,520 (\$13,939), while PBT was over triple the cost at €39,610 (\$47,928); but without toxicity analysis, definitive conclusions remain elusive.

A Markov report from Holland²⁹ specifically examined patients with stage III-IV oral cavity, laryngeal, and pharyngeal cancers. Cohorts were divided into three groups, IMRT for all patients, intensity-modulated proton therapy (IMPT) for all patients, and mixed IMPT/IMRT with IMPT only if “expected to be cost-effective” (calculated based on estimated 6-month-risk of xerostomia). Radiation Therapy Oncology Group (RTOG) grades of normal tissue complications were specifically used to define toxicities. At 12 months, xerostomia and dysphagia rates were 22 and 18% respectively with IMPT, 36 and 21% with mixed IMPT/IMRT, and 44 and 23% with IMRT. Though all three groups had similar QALYs (6.52 IMRT, 6.62 IMPT, 6.56 mixed), costs were €50,989 (\$61,697) for IMPT, €41,038 (\$49,656) for IMRT, and €43,650 (\$52,816) for mixed. This is the strongest evidence to date suggesting that select head/neck cancer patients may have decreased side effects while also being treated with nearly the same radiotherapy costs as standard-of-care IMRT. However, lack of clinical data continue to hamper interpretations from this study; utility scores were based on a poorly-performed cross-sectional analysis, and probabilities of disease progression were also based on outdated studies. Nevertheless, despite incomplete characterization of toxicities (e.g. need for gastrostomy tubes), this report remains the only study to date that has shown a CE benefit to PBT in head/neck cancers. Therefore, further characterizing

the population of head/neck cancer patients most benefitting from PBT is of great clinical interest not only for future toxicity analyses, but also for CE analyses. These data are depicted in Table 4.

Recent clinical data in oropharyngeal and nasopharyngeal cancer, utilizing matched-cohort (PBT versus IMRT) methodology, demonstrate an approximate 50% reduction in the use of gastrostomy feeding tubes for the PBT cohort.³⁰⁻³¹ PBT's potential for improving CE and QALYs for this event would be expected to be considerable; a phase II trial in oropharyngeal cancer is currently underway. This group has recently conducted a modeling study to evaluate episodic costs of care including management of acute toxicities in head/neck cancer, and the data favorably support PBT.³² Therefore, *if prospective studies confirm a meaningful improvement in long-term outcomes with PBT, the cost-effectiveness of PBT may reach an acceptable threshold for more routine implementation in select head and neck cancer cases.* Though more subgroups could economically benefit most from PBT, attention also must be paid to those patients that are likely not to derive this benefit, in order to develop a more complete and balanced economic assessment.

Pediatric Cancers

Table 5 summarizes data for PBT CE in pediatric cancers. The aforementioned Swedish study assessed CE of PBT for medulloblastoma, albeit being performed prior to dose de-escalation.¹⁸ Using Markov modeling, a cohort of children (5-years-old) with medulloblastoma was assessed, comparing PBT or intensity-modulated radiotherapy (IMRT). Though initial costs of PBT were estimated as €10,218 (\$12,364) compared with €4,239 (\$5,129) for conventional radiotherapy (2.4-fold increase), total costs of adverse effects were estimated at €4,232 (\$5,121) and €33,857 (\$40,967) respectively (8-fold difference in favor of PBT), yielding total costs of €14,450 (\$17,484) and €38,096 (\$46,096) respectively (2.6-fold decrease with PBT). Analysis revealed that the greatest factors contributing to adverse event

costs were IQ, hearing loss, and growth hormone deficiency. QALYs were similar in both groups, 12.778 and 12.095, respectively. The same group performed another Markov analysis in medulloblastoma³³ that illustrated PBT domination of cost analyses (€23,647 (\$28,613)/patient saved; 0.683 QALYs gained from PBT). However, a flaw in this study limiting reliability was the generalization of PBT costs per fraction as being similar for all types of cancer (regardless of treatment time and other parameters).

In a similar study of medulloblastoma with Monte Carlo methodology, 5-year-old children received PBT or IMRT.³⁴ Whereas lifetime (including morbidity management) IMRT cost was estimated at \$112,790 (€100,755), PBT was \$80,211 (€71,652); total QALYs favored PBT-treated patients, 17.37 versus 13.91. Because the study was designed to assess costs throughout lifetime, post-treatment costs were not tracked until the children reached 18 years of age. Though not consistent with the previous report in terms of QALY (attributed to different actual costs of IQ loss), this report corroborated that decrease in adverse effects from PBT offered cost benefits.

Another report specifically examining growth hormone deficiency in pediatric brain tumor patients³⁵ used Markov simulation to compare PBT and IMRT, and concluded that PBT's advantages are maintained across an entire dose-range, with greater differences in hypothalamic dose between photons and protons creating larger CE spreads. Similar results were obtained by Hirano and colleagues when examining only hearing loss due to cochlear dose reduction with PBT versus IMRT.³⁶ Applicability of both these data are limited owing to lack of assessing costs for all potential major adverse effects.

Emerging data suggest dosimetric superiority and/or reduced toxicities with PBT for several pediatric tumors: retinoblastoma³⁷, brain tumors and craniopharyngiomas³⁸⁻⁴¹, rhabdomyosarcoma⁴², ependymoma⁴³, and esthesioneuroblastoma.⁴⁴ However, CE analyses for these have not been performed.

Esophageal Cancer

Though data examining efficacy and favorable dosimetry of PBT for esophageal cancers have been published/presented, there are currently no data examining CE in esophageal cancer. Data from a multi-institutional retrospective cohort of neoadjuvant chemoradiotherapy followed by resection in 582 patients, treated with photons (n=471) or protons (n=111) revealed that the post-operative hospital length of stay was shorter for proton patients (9 vs. 12 days, $p<0.0001$) – largely attributable to fewer cardiopulmonary and wound toxicities. Ninety-day mortality rates were also lower for PBT. A more detailed analysis of this dataset is currently underway, including a CE analysis.⁴⁵

Skull Base Cancers

The previously discussed Dutch non-modeling analysis¹⁷ also analyzed patients with skull base chordomas. Conventionally fractionated stereotactic RT for chordomas cost an estimated €13,970 (\$16,904), as compared with €30,530 (\$36,941) for PBT; no toxicities were discussed. Because skull base tumors are currently a relatively agreed-upon indication for PBT, and this study was only a “cost” study that failed to consider decreases in toxicity with PBT, CE studies are clearly needed. These data are presented in Table 6.

Uveal Melanoma

Markov-modeled results were recently published from Mayo Clinic⁴⁶ comparing PBT, enucleation, and plaque brachytherapy. The authors demonstrated similar costs of \$22,772 (€20,342) for enucleation, \$24,894 (€22,238) for PBT, and \$28,662 (€25,604) for plaque brachytherapy. QALYs were nearly identical at 2.918, 2.938, and 2.994 years, respectively. There were wide numerical

variations based on estimates seen on one-way sensitivity analysis, and a dubious model assumption was that all recurrences necessarily led to metastases or death. The analysis did not stratify tumors for size, and also did not use recent data that shows PBT having superior survival as compared to enucleation, and improved vision preservation compared to plaque brachytherapy.⁴⁷⁻⁴⁸ Improved outcomes with PBT against brachytherapy in phase III trials are emerging⁴⁹⁻⁵⁰; future CE analyses could correspondingly be more favorable.⁴⁸ Results are summarized in Table 6.

Critiques and Applicability

Foremost, it is important to mention that there is no “perfect” CE analysis. It is near impossible to account for every salient logistical and economic variable in modeling; examples include initial capital investment of PBT facility (real estate/construction), hours of operation, worker labor costs, adjustment for monetary value over time, taxes, interest rates, treatment time per patient, patient commuting costs, and machine maintenance. Moreover, very few studies use prospectively-collected data as the basis for utilities, costs, and other assumptions, and this can be a major detriment to the quality of available CE analyses. Additionally, no study evaluated herein took into account all commonly encountered major adverse events, including secondary malignancies (though less important for poor-prognosis tumors). Furthermore, costs of adverse effects (and worsening of preexisting comorbidities) are substantially uncertain owing to the enormous possibilities of treatments/procedures (and physician visits) to ameliorate a given adverse effect. Hence, because no CE study can assess all of the aforementioned parameters, it is important for readers to critically evaluate the scope of included variables in each study in order to determine practical applicability.

There are several other factors to be discussed with regard to applicability. Though CE analyses rely heavily on established clinical data and an unavoidable assortment of assumptions, older CE reports

(e.g. Lundkvist et al.¹⁸) often rely on studies with outdated forms of radiotherapy, techniques, and technologies. Often in CE reports, the time point of CE calculations after therapy is ambiguous; though there is often adjustment for the value of money over time (some studies do not perform this²¹), some studies only adjust select variables for this factor. Furthermore, different healthcare systems in other countries can have vastly different costs of goods/services as well as WTP thresholds (e.g., Grutters et al.²⁷), thus leading to potentially different economic conclusions.

These conditions as well as others have been postulated by the Panel for Cost-Effectiveness in Health and Medicine as necessary for high-quality and reliable CE analyses.⁵¹ Other discussion points set forth include the necessary inclusion of quality-of-life data (of which there is very little for PBT to date, although many ongoing clinical trials are testing this), because patients may not be in “perfect-quality” of health for any QALYs calculated. Though the panel also recommends that high-quality and diverse data be used as the basis of study assumptions, these clearly do not exist at this time for PBT (e.g. prospective PBT publications). There are also controversies regarding whether costs of pre-existing diseases that could be impacted by RT should be added (e.g. COPD exacerbation in post-RT lung cancer patients). Next, though the panel recommended a 3% discount on costs to adjust for the value of money, this may not be applicable in other countries, who often adjust them differently (e.g. Grutters et al.²⁷). Lastly, it is also noted that very few studies account for costs associated with mortality, and no paper in this review did so. Taken together, according to these guidelines, it is clear that on multiple counts, no study in this review meets appropriate criteria set forth in this publication. Nevertheless, despite the overall low level of evidence, we suggest that a more reliable method of ascertaining applicability of CE studies is to evaluate corroboration between results and conclusions of repeated CE or even “cost-only” studies (e.g. prostate and pediatric brain cancer CE studies). Additionally, we encourage that readers use this overall low quality of evidence to holistically assess which tumors could

be good candidates for further CE study. Replicative studies are greatly encouraged to validate claims and conclusions.

DISCUSSION AND CONCLUSIONS

Greater availability of PBT has necessitated the generation of a larger body of clinical data to support its efficacy and utility. Not only are clinical data needed, economic analyses are also crucial; these are largely dependent on currently-sparse outcomes research, albeit rapidly changing. We reviewed all known data of CE analyses of PBT, and though the paucity of data permits few robust conclusions, no previous summative reviews are currently in existence in order to guide medico-economic decisions.

Clinical utilization of PBT has been limited to date; capacity constraints have precluded the conduct of large scale trials which are required to evaluate the comparative effectiveness of PBT versus conventional RT modalities.⁵² Meanwhile, because of increased up-front investments and operational costs, initial costs of PBT are higher; these must be contextualized by recognizing that within the last year or so, smaller, single room solutions have become available. These reduce both up-front and some operational costs, implying that going forward, the ratios generated for CE comparisons will be relatively improved for PBT, as compared to the historical data presented herein.

There are several additional limitations of the data analyzed in this review, including the need to utilize caution when interpreting any results based on several assumptions (including scope of costs encompassed in each study, not limited to initial capital investment, labor, and operational/maintenance costs which were included in only one paper¹⁷) made in modeling, but also because some conclusions are from non-modeling studies (e.g. Medicare estimates) or theoretical/hypothetical studies that were not explicitly discussed in this paper.⁵³⁻⁵⁴ Furthermore,

different healthcare systems can have radically different estimates of costs, and applicability to health economics in other countries can be predictably limited. Additionally, time points of CE measurement (as designated by authors) after treatment can have large impacts on QALY and ICER values and hence the conclusions of a particular analysis. Moreover, even newer methods to model CE are also emerging⁵⁵ that necessitate cross-corroboration of data between different methods in order to remain confident in data, results, and conclusions.

The arbitrary WTP threshold of \$50,000 (€44,665)/QALY has been heavily criticized as a historical remnant, and many have argued for increases, especially in light of inflation and newer economic times.⁵⁶ Studies generally use thresholds from \$20,000 (€17,866) to \$100,000 (€89,330) per QALY; however, per a review of the Tufts CE Registry, though a majority of studies still use the so-called “historical threshold”, \$100,000/QALY as well as both \$50,000/QALY and \$100,000/QALY are more frequently being utilized.⁵⁷ Complicating matters is that WTP depends on perspective (patient/practitioner/payer): type of insurance company, specific clinical circumstance of the patient, “clinical aggressiveness” of the managing physician, and location of the particular market (competing facilities/referral patterns). There are several other factors that influence this parameter, including the country/society/economic system, noteworthy socioeconomic events therein, and several other factors even involving the global economy. Though having similar values aids in study standardization, this is misleading secondary to a number of other nonstandardized variables between studies. Perhaps, therefore, by integrating and prioritizing these variables, it is most appropriate to create several different WTP thresholds that can – and should – fluctuate with time.

Available evidence suggests PBT for unselected populations with early stage lung cancer and prostate cancer is economically suboptimal. It is crucial to recognize that historical PBT approaches for prostate cancer have utilized opposed lateral fields, which therefore do not spare the anterior rectal wall; recent developments, such as the availability of rectal gel spacers, and the ability to perform *in vivo*

Bragg-peak range verification would allow the use of anterior or anterior-oblique PBT beam arrangements which would be substantially superior from a dosimetric perspective and could hence improve PBT's CE profile, albeit to an unknown degree.⁵⁸⁻⁵⁹ Conversely, PBT is likely the economic standard of care for a significant proportion of pediatric patients, although CE evidence only exists for pediatric brain tumors. Many other cancers with the potential for significant radiation-induced toxicities such as head/neck cancers, esophageal cancer, advanced non-small cell lung cancer, brain tumors, skull base neoplasms, ocular cancers, and certain left-sided breast cancers as well as certain soft tissue sarcomas are all potential candidates for cost-effective PBT. Data for these indications, and also for re-irradiation in multiple anatomic sites are rapidly emerging, and will likely result in radical revision of many conclusions that can be made currently.

The ability of PBT to reduce cardiopulmonary morbidity cannot be overlooked. Darby et al.⁶⁰ demonstrated a linear relationship between mean heart dose and cardiac morbidity, with greatest risk among patients with baseline cardiac risk factors. The ability of PBT to effectively reduce cardiac doses in left-sided breast cancer patients⁶¹ may therefore lead to decreased cardiac morbidity and mortality, although increased experience and longer follow-up times with PBT are necessary for corroboration. This suggests that PBT can yield improved CE if certain high-risk subsets of breast cancer patients are selected.

Similarly, data are needed for patients with advanced lung cancer in whom cardiopulmonary tolerance doses have a major impact on outcome, including survival, and they represent a patient population in whom the potential for toxicity-related medical costs also increase. RTOG 0617 demonstrated inferior survival in the higher dose arm, linked to probable cardiopulmonary toxicities associated with higher dose.⁶² Multivariate analysis for factors associated with overall survival between standard and high-dose radiation, demonstrated lower heart V5 as being associated with improved overall survival. More recent prospective data demonstrate that heart doses with PBT are substantially

lower compared to IMRT. The heart V5 value for 103 patients on an ongoing prospective trial revealed mean values for heart V5 of 14.6% ($\pm 13.6\%$ SD) with PBT, compared to 44.3% ($\pm 33.3\%$ SD) with IMRT ($p < .0001$).⁶³ Liao and colleagues previously investigated whether heart dose affects overall survival in NSCLC by evaluating the relationship between mean heart dose and overall survival in a 532 patient retrospective cohort.⁶⁴ Mean heart doses ranged from 0 to 55.4 Gy, and the median was 22.3 Gy, 15.1 Gy and 6.5 Gy for 3DCRT, IMRT, and PBT, respectively. On multivariate analysis, mean heart doses above the 25th percentile were significantly associated with increased risk of death. To evaluate the impact of PBT further, these investigators analyzed the mean heart dose in 103 patients randomized to a 2-institution PBT versus RT trial, and found that mean heart doses have been consistently lower for PBT.⁶⁵ These important data have now been used to develop the randomized RTOG 1308 trial, testing RT versus PBT, and the clear dosimetric advantages with putative cardiopulmonary toxicity reductions will likely have a major impact on future CE analysis for NSCLC.

Dramatic technological innovations to optimize PBT are occurring at this point, including pencil beam scanning, intensity modulated proton therapy, image guidance, utilization of hypofractionation⁶⁶, and development of compact units; these will likely further decrease treatment costs.⁶⁷ The use of cyclotrons versus synchrotrons has been known to be related to CE⁶⁸, along with postulation that four-dimensional proton treatment may one day be the standard of care for some tumors.⁶⁹ Recent advances in optimizing treatment times⁷⁰, proton units⁷¹, beam energies⁷², and field design⁷³ can improve CE, as can finding a balance of PBT-indicated cases and “nonessential” cases.⁷⁴ None of these factors were taken into account by any study in this review. However, partially due to these innovations, data state that over the next decade, treatment costs could drop by a very substantial 20%.⁷⁵ Moreover, PBT reimbursements already have decreased from the past as well.⁷⁶ Lastly, because clinical and toxicity data are currently accruing and maturing for PBT, a clearer picture of efficacy and thus economic balances

will likely be available in the future. This indicates that the present review may likely need to be revised once further data become available.

In summary, it is highly unlikely that PBT will be the most economical option for all cancers, or even all patients for a given type of cancer. Rather, the major goal for ongoing and future research will be to identify the subpopulation(s) of each cancer type for whom PBT is most cost-effective. It is also very important to consider that CE analyses are inherently highly dependent on existing (largely retrospective) literature, study methodology, and a multitude of variables that cannot all be adequately captured on analysis. Hence, we encourage reporting of CE data from a variety of sources in order to be able to corroborate the findings in this review.

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FIGURE LEGENDS

Figure 1. PRISMA diagram illustrating systematic searches for this review.

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Reference, Country, Year of Cost Analysis	Methodology	Key Assumptions of Model	Cancer Details	Therapy Comparisons	Total Costs	QALYs	ICER	Conclusions & Criticisms
Lundkvist, Sweden ¹⁸ 2002	Markov	Adverse events were GU and GI; costs based on “standard treatments” without management physician costs PBT has 60% the risk of adverse events as CRT No capital investment, labor, or operational costs included	65-year-old men with prostate cancer undergoing PBT or 35 fractions CRT	PBT vs. CRT	€7,952 (\$8,902) higher for PBT	0.297 gained from PBT	€26,800 (\$30,001)/QALY	PBT not cost-effective Yearly excess mortality rate of 2.5% due to prostate cancer used for all patients uniformly Assumed that PBT would “be delivered at a higher target radiation dose than conventional radiation, and thereby increases the tumour control” with arbitrary 20% improvement in tumor control and 20% reduction in mortality with PBT; no RT dose relationship given Assumed that 300 prostate patients treated per year (same as the number of head/neck patients) Adverse event of sexual dysfunction not noted Sensitivity analysis: appropriate variability in costs; PBT with increased costs at all examined levels
Konski, USA ¹⁹ 2005	Markov	Post-treatment health states include progression responsive (or lack thereof) to hormones Assumed PBT could increase dose by 10 Gy to 91 Gy safely with 10% decrease in biochemical failure QOL analysis incorporated into model No capital investment, labor, or operational costs included	Intermediate risk patients either 60 or 70 years of age	PBT vs. IMRT	70-year-old: \$63,511 (€56,734) PBT \$36,808 (€32,880) IMRT 60-year-old: \$64,989 (€58,054) PBT \$39,355 (€35,156) IMRT	9.91 9.45	\$63,578 (€56,794)/QALY for 70-year-old \$55,726 (€49,780) for 60-year-old	PBT not cost-effective as IMRT, but could be for young patients Did not directly deal with toxicity/adverse effects in Markov model (only survival) Relationships between dose and freedom from biochemical failure obtained largely from own institutional analyses Authors acknowledge that essentially all conclusions rely on whether extra 10 Gy could be delivered; utility values remained the same between IMRT and PBT (owing to dearth of data) Arbitrary assumption that all hormone-refractory patients die in 1 year Sensitivity analysis: appropriate variability in costs; PBT with increased costs unless its costs at \$39,000 (€36,744) or below
Peeters, Holland ¹⁷ 2007	Cost estimates from literature	Assumed facilities of either 2-room photons, 3-room PBT, or PBT/carbon ion (3 rooms) with average lifetime 30 years Investment capital, operational, and labor costs from literature and existing business plans, including interest and replacing linacs every 10 years	No stratification for risk category of prostate cancer	PBT vs. IMRT	€16,090 (\$19,469) PBT, €18,160 (\$21,974) IMRT	-	-	PBT superior due to decrease in adverse effects Compared with photon facility, PBT costs increase by 3.2 and particle facility by 4.8 Assumed linear relationship between cost/fraction and number of fractions Lack of clarity on types of tumors treated at certain frequencies (or lack thereof) Not a “cost-effectiveness” analysis, without assessment of outcomes or toxicities Sensitivity analysis: performed for operational costs and not cancer specifically; appropriate variability in costs; treatment timings/capacity potentially most variable

1	Aizer, USA ²⁰	Medicare & SEER data	Survival per Social Security Life Table	N = 21 subset of low-risk patients treated with PBT	PBT vs. IMRT vs. WW vs. surgery vs. cryotherapy vs. SBRT vs. brachytherapy vs. ADT	\$42,772 (€38,208) PBT \$29,616 (€26,456) IMRT \$2,766 (€2,471) WW ~\$14,000 (€12,506) surgery \$12,516 (€11,180) cryotherapy \$26,930 (€24,056) SBRT \$16,883 (€15,082) brachytherapy \$7,070 (€6,316) ADT	-	-	PBT not cost-effective in low-risk disease Applicability greatly limited, only to Medicare patients and with very low sample size for PBT Only a small “cost” (not “cost-effectiveness”) analysis with essentially no accountability for outcomes and toxicities between groups, especially PBT group Sensitivity analysis: none
2	2010		Cost per Medicare payments for inpatient/outpatient services with diagnosis as prostate cancer						
3			No patients on hormones						
4			No capital investment, labor, or operational costs included						
5	Parthan, USA ²¹	Markov	Toxicities included combinations of GI, GU, and sexual dysfunction using RTOG scale	65-year-old patients either eligible for radiotherapy or declining surgery	PBT vs. IMRT vs. SBRT	\$69,412 (€62,005) PBT \$33,068 (€29,540) IMRT \$24,873 (€22,219) SBRT	8.06 PBT 8.05 IMRT 8.11 SBRT	\$13,755,207 (€12,287,472)/QALY for PBT over SBRT	PBT not cost-effective. From payer perspective, SBRT cost-effective in 75% and 94% of simulations vs. PBT (threshold \$50,000/QALY). Assumed all toxicities associated with 1-time cost without reassessment thereafter No discount rate (changes in value of money) with time Sensitivity analysis: SBRT dominant in cost-effectiveness, including 94-95% chance of being more cost-effective vs. PBT; though one-way sensitivity analysis results acceptable, did not perform the more reliable probabilistic multivariate analyses Little mention of doses for any modality, or dose-dependent toxicities
6	2011		Late toxicities defined as 12+ months post-therapy						
7			Outcomes & mortality from literature & SEER						
8			No capital investment, labor, or operational costs included						
9	Yu, USA ²²	Medicare estimates	Toxicities included GI, GU, and other	27,647 patients aged 66-94 undergoing prostate radiation; 27,094 IMRT and 553 PBT	PBT vs. IMRT	\$32,428 (€28,968) PBT \$18,575 (€16,593) IMRT	-	-	PBT not cost-effective. PBT group more likely to be younger, Caucasian, wealthier, home very close or very far from facility. At 12 months, no differences in GU (17.5% IMRT, 18.8% PBT), GI (10.2%, 9.9%), other (5.6%, 4.5%) toxicities Did not examine outcomes (only toxicity) Only applicable to Medicare patients Only a “cost” analysis with no “cost-effectiveness” measures Lack of accountability at specific disease states (e.g. recurrence) or RT doses Sensitivity analysis: none
10	2008-2009		Matched PBT and IMRT patients for equitability						
11			No capital investment, labor, or operational costs included						
12	Boyal, USA ²³	SEER and Medicare data	Equal disease control with PBT and IMRT	Age 66+ patients	PBT vs. IMRT	-	-	-	PBT not cost-effective. PBT needs to have 41% reduction in overall toxicity to be considered cost-effective (threshold \$100,000/QALY). Currently in abstract form without precise methodologies reported; unclear if quality of life included Sensitivity analysis: quality of analysis unverifiable; relative increase in risk of composite toxicity needs to be 40-80% in order for PBT to be cost-effective at the \$100,000 (€89,330)/QALY level
13	2011		Utility values based on Medicare and literature						
14			No capital investment, labor, or operational costs included						

Table 1. Summaries of available studies examining cost-effectiveness for prostate tumors. SEER, surveillance, epidemiology, and end-results; PBT, proton beam radiotherapy; IMRT, intensity-modulated radiation therapy; IMRT, intensity-modulated radiation therapy; SBRT, stereotactic body radiation therapy; WW, watchful waiting; ADT, androgen deprivation therapy; QALYs, quality-adjusted life-years; ICER, incremental cost effectiveness ratio; GU, genitourinary; GI, gastrointestinal; RTOG, Radiation Therapy Oncology Group.

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Reference, Country, Year of Cost Analysis	Methodology	Key Assumptions of Model	Cancer Details	Therapy Comparisons	Total Costs	QALYs	ICER	Conclusions & Criticisms
Lundkvist, Sweden ²⁴ 2002	Markov	Patients at risk of all-cause death every year; tumor-related death in first 10 years Mortality rates per Swedish life tables; population risk for cardiac disease based on national registry based on age Did not count regional nodal irradiation patients and hence assumed 10-year survival of 65% No capital investment, labor, or operational costs included	55-year-old women with left-sided breast cancer, secondary analysis of subpopulation assumed to have double the cardiac disease risk	PBT vs. WBI	€11,248 (\$13,610) PBT €5,001 (\$6,051) IMRT	12.35 PBT 12.25 IMRT	€66,608 (\$80,596)/QALY for general patients, €34,290 (\$41,491)/QALY for subpopulation	PBT potentially economically beneficial in select patients at high risk of cardiac toxicity Arbitrary assumptions of 14% pneumonitis risk and 75% of those patients would take indeterminate amount of sick leave from work (productivity loss) Simulated younger 55-year-old patients despite average 63-year-old Swedish breast cancer patient, but many costs/utilities were based on national averages Sensitivity analysis: appropriate variability in costs; variations in estimated risk of cardiac disease most related to changes in costs of PBT vs. WBI
Lundkvist, Sweden ¹⁸ 2002	Markov	Patients at risk of all-cause death every year; tumor-related death in first 10 years Cardiac diseases categorized as fatal (death-inducing) or nonfatal Base incidences of cardiac disease were age/gender-matched No capital investment, labor, or operational costs included	55-year-old women with left-sided breast cancer, PBT vs. CRT in 25 fractions	PBT vs. CRT	€5,920 (\$7,163) higher for PBT	0.17 gained from PBT	€34,200 (\$30,551)/QALY	PBT not cost-effective Used now-outdated RT schemas, including no breast boost and assumption of 14% risk of severe pneumonitis, no corresponding utility reductions Adverse events included only cardiopulmonary toxicities and no others, including second cancers Little data for risk of cardiac events after breast cancer RT Sensitivity analysis: wide variability in costs; cardiac risk reduction most related to variability in costs
Taghian, USA ²⁵ 2003	Medicare estimates	Costs (professional / technical) assumed to be for 25 fractions with 5-fraction boost without chemo/hormonal therapy No capital investment, labor, or operational costs included	Breast cancer	PBT vs. WBI (50 Gy in 25 fractions + 10 Gy in 5 fractions) vs. mixed photon-electrons (opposed lateral photons, en face electrons)	\$13,200 (€11,792) PBT \$10,600 (€9,469) WBI \$5,300 (€4,734) mixed	-	-	PBT only modestly more cost-effective than conventional WBI “Cost” analysis and not a “cost-effectiveness” analysis without comparison for outcomes, toxicities, etc. Only applicable to USA Medicare patients owing to the specific source of cost data Sensitivity analysis: none

Ovalle, USA ²⁶	Medicare estimates	Costs strictly based on Medicare figures without other salient parameters	Early-stage breast cancer patients suitable for APBI	APBI vs. WBI APBI using SAVI, MammoSite, linear accelerator-based 3DCRT WBI using 3DFIF, hypofractionation, IMRT, hypofractionated IMRT	\$13,883 (€12,402) PBT APBI \$14,859 (€13,265) SAVI APBI \$12,245 (€10,938) MammoSite APBI \$6,771 (€6,049) 3DCRT APBI \$13,149 (€11,746) 3DFIF WBI \$10,070 (€8,995) Hypofractionated WBI \$19,599 (€17,508) IMRT WBI \$11,747 (€10,494) Hypofractionated IMRT WBI	-	-	Technical components of treatment provide most sources of cost variations
2015		No capital investment, labor, or operational costs included						Currently in abstract form without precise methodologies reported; also used Medicare estimates without data on several other parameters as above
								Sensitivity analysis: none

Table 2. Summaries of available studies examining cost-effectiveness for breast tumors. NSCLC, non-small cell lung cancer; PBT, proton beam radiotherapy; WBI, whole breast irradiation; CRT, conventional radiotherapy; SRS, stereotactic radiosurgery; SBRT, stereotactic body radiation therapy; QALYs, quality-adjusted life-years; ICER, incremental cost effectiveness ratio; IMRT, intensity-modulated radiation therapy; APBI, accelerated partial breast irradiation; SAVI, strut-adjusted volume implant; 3DCRT, 3-dimensional conformal radiotherapy; 3DFIF, 3-dimensional field-in-field.

Reference, Country, Year of Cost Analysis	Methodology	Key Assumptions of Model	Cancer Details	Therapy Comparisons	Total Costs (95% CI)	QALYs (95% CI)	ICER	Conclusions & Criticisms
Grutters, Holland ²⁷ 2007	Markov	Health states included grade 3+ RT pneumonitis or esophagitis with corresponding quality of life; second malignancies or cardiac events not counted Estimates of nearly all event probabilities based on meta-analysis Discounting of future effects and costs performed at 1.5% and 4% respectively (within Dutch context) Incomplete operational costs; no capital investment or labor costs included	Inoperable stage I NSCLC Operable stage I NSCLC	PBT vs. CIRT vs. CRT vs. SBRT CIRT vs. SBRT	€27,567 (€22,713-€32,368) (\$33,356; \$25,421-\$36,228) PBT €19,215 (€14,273-€27,415) (\$23,250; \$15,975-\$30,684) CIRT €22,696 (€19,566-€26,274) (\$27,462; \$21,904-\$29,407) CRT €13,871 (€11,124-€17,413) (\$16,784; \$12,451-\$19,494) SBRT €8,485 (€6,497-€11,613) (\$9,499) SBRT €14,620 (€10,231-€22,694) (\$16,366) CIRT	2.33 (1.96-2.74) PBT 2.67 (2.28-3.04) CIRT 1.98 (1.78-2.20) CRT 2.59 (2.31-2.88) SBRT 3.20 (3.03-3.36) SBRT 3.16 (2.86-3.42) CIRT	€67,257 (\$75,291)/QALY for CIRT vs. SBRT	SBRT and CIRT most cost-effective for inoperable cases, SBRT most for operable cases Only used a time frame of 5 years post-treatment (owing to no data thereafter) Limited data available on operable stage I patients (128 SBRT and 31 CIRT patients from 2 published studies) Utility numbers for acute effects equated to those after treatment (owing to lack of direct data) Nearly all studies in the one meta-analysis used to derive event probabilities were noncomparative, single-arm observational studies No costs/utilities for cancer recurrence Used payment threshold of €80,000/QALY Sensitivity analysis: appropriate variability in costs; PBT with increased costs at all examined levels
Peeters, Holland ¹⁷ 2007	Cost estimates from literature	Assumed facilities of either 2-room photons, 3-room PBT, or PBT/carbon ion (3 rooms) with average lifetime 30 years Investment capital, operational, and labor costs from literature and existing business plans, including interest and replacing linacs every 10 years	Inoperable stage I NSCLC	PBT vs. SBRT vs. 3DCRT	€12,380 (\$13,851) PBT €3,720 (\$4,501) SBRT €8,150 (\$9,862) 3DCRT	-	-	PBT not cost-effective Compared with photon facility, PBT costs increase by 3.2 and particle facility by 4.8 Assumed linear relationship between cost/fraction and number of fractions Lack of clarity on types of tumors treated at certain frequencies (or lack thereof) Not a “cost-effectiveness” analysis, without assessment of outcomes or toxicities Sensitivity analysis: performed for operational costs and not cancer specifically; appropriate variability in costs; treatment timings/capacity potentially most variable
Lievens, Belgium ²⁸ 2012	Markov	Defined for 10 years with disease states of controlled disease, locoregional or distant progression, and death No capital investment, labor, or operational costs included	Locally advanced NSCLC, concurrent chemotherapy	PBT vs. IMRT vs. 3DCRT	PBT with €18,875 (\$21,130) increase from 3DCRT, and €14,257 (\$15,960) increase from IMRT	PBT with 0.664 QALY increase from 3DCRT, and 0.549 increase from IMRT	PBT with €34,396 (\$38,505)/QALY from 3DCRT, and €31,541 (\$35,309)/QALY from IMRT	PBT is “borderline cost effective in the Belgian health care context” Currently in abstract form without precise methodologies reported including chemotherapy details, costs, and complications/adverse effects Sensitivity analysis: none

Table 3. Summaries of available studies examining cost-effectiveness for lung tumors. NSCLC, non-small cell lung cancer; PBT, proton beam radiotherapy; CIRT, carbon ion radiation therapy; CRT, conventional radiotherapy; SBRT, stereotactic body radiation therapy; IMRT, intensity-modulated radiation therapy; 3DCRT, three-dimensional conformal radiation therapy; QALYs, quality-adjusted life-years; CI, confidence interval; ICER, incremental cost effectiveness ratio.

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Reference Country, Year of Cost Analysis	Methodology	Key Assumptions of Model	Cancer Details	Therapy Comparisons	Total Costs	QALYs	ICER	Conclusions & Criticisms
Lundkvist, Sweden ¹⁸ 2002	Markov	Mortality data for first 9 years from national registry; thereafter, assumed to have normal age-specific mortality Used constant utility score from quality of life studies No capital investment, labor, or operational costs included	65-year-old patients with head and neck cancers treated with PBT vs. 35 fractions CRT (including hyperfractionation)	PBT vs. CRT	€3,887 (\$4,703) higher for PBT	1.02 gained from PBT	€3,800 (\$4,254)/QALY	PBT can potentially be cost-effective, especially in light of side effects Questionable utilization of hyperfractionation; data used but not incorporated into costs Calculated constant dentistry costs, unclear on relationship to dose/fractionation/modality No data on toxicities/quality of life after PBT in existence for use at the time, thus inherently inaccurate comparison IMRT largely not used for these cancers at time of publication, limiting toxicity data Sensitivity analysis: appropriate variability in costs; PBT with increased costs at all examined levels
Peeters, Holland ¹⁷ 2007	Cost estimates from literature	Assumed facilities of either 2-room photons, 3-room PBT, or PBT/carbon ion (3 rooms) with average lifetime 30 years Investment capital, operational, and labor costs from literature and existing business plans, including interest and replacing linacs every 10 years	No stratification for type or stage of head and neck cancer	PBT vs. IMRT	€39,610 (\$47,928) PBT, €11,520 (\$13,939) IMRT	-	-	PBT not cost-effective Compared with photon facility, PBT costs increase by 4.2 and particle facility by 4.8 Assumed linear relationship between cost/fraction and number of fractions Lack of clarity on types of tumors treated at certain frequencies (or lack thereof) Not a “cost-effectiveness” analysis, without assessment of outcomes or toxicities Sensitivity analysis: performed for operational costs and not cancer specifically; appropriate variability in costs; treatment timings/capacity potentially most variable
Raemakers, Holland ²⁹ 2010	Markov	Many diverse health states, depending on disease status and RTOG grade 2+ dysphagia and/or xerostomia IMPT “if efficient” (mixed) group calculated based on ICERs on case-by-case basis and probability of xerostomia (mean probability of this group 37% vs. 26% IMPT and 45% IMRT) Assumed that toxicities within first 6 months were partly reversible but irreversible thereafter No capital investment, labor, or operational costs included	Stage III/IV oral cavity, laryngeal, pharyngeal	IMPT vs. IMRT vs. mixed	€50,989 (\$61,697) IMPT €41,038 (\$49,656) IMRT €43,650 (\$52,816) mixed	6.62 IMPT 6.52 IMRT 6.56 mixed	Mixed vs. IMRT €60,278 (\$67,478)/QALY, IMPT vs. mixed €127,946 (\$143,229)/QALY	12-month xerostomia/dysphagia rates 22%/18% IMPT, 36%/21% mixed, 44%/23% IMRT Individual calculation of cost-effectiveness and toxicity risk very important to determine optimal modality Disease progression statistics based on old study with CRT Utility scores for disease states based upon relatively weak cross-sectional analysis Vague methodology on time course and frequency of time points used to assess toxicities Not all toxicities and costs assessed (e.g. odynophagia requiring pain medications, gastrostomy tube, etc.) Sensitivity analysis: appropriate variability in costs; IMPT-only with increased costs at all examined levels, but was only compared to IMRT-only (mixed group not analyzed)

Table 4. Summaries of available studies examining cost-effectiveness for head and neck tumors. NSCLC, non-small cell lung cancer; PBT, proton beam radiotherapy; CRT, conventional radiotherapy; CSRT, conventionally-fractionated stereotactic radiation therapy; SBRT, stereotactic body radiation therapy; IMRT, intensity-modulated radiation therapy; 3DCRT, three-dimensional conformal radiation therapy; IMPT, intensity-modulated proton therapy; QALYs, quality-adjusted life-years; ICER, incremental cost effectiveness ratio.

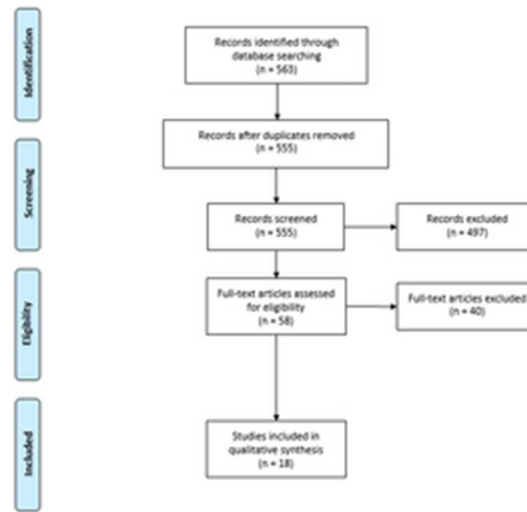
Reference, Country, Year of Cost Analysis	Methodology	Key Assumptions of Model	Cancer Details	Therapy Comparisons	Total Costs	QALYs	ICER	Conclusions & Criticisms
Lundkvist, Sweden ¹⁸ 2002	Markov	25% of IQ loss attributable to RT 75% of patients with hearing loss incur only one-time cost of hearing aid Capital investment costs distributed evenly among all patients treated at facility; operational/labor costs not included	5 year old children with medulloblastoma	PBT vs. IMRT	€14,450 (\$17,484) PBT €38,096 (\$46,096) IMRT	12.778 PBT 12.095 IMRT	-	PBT superior due to less IQ loss, hearing loss, growth hormone deficiency Used same utility values for pediatric and adult life Did not account for QOL secondary to IQ and hearing loss Little mention of RT doses, was performed prior to dose de-escalation to neuraxis Sensitivity analysis: appropriate variability in costs; IMRT with increased costs at all examined levels
Lundkvist, Sweden ³³ 2002	Markov	25% of IQ loss attributable to RT Average IQ loss 17 points Risk of hearing loss around 13% No capital investment, labor, or operational costs included	5 year old children with medulloblastoma treated with PBT or 25 fractions CRT	PBT vs. CRT	€23,647 (\$28,613) lower for PBT	0.683 gained from PBT	€-23,600 (-\$26,419)/QALY	PBT superior Costs per fraction of RT used for all types of cancers (including palliation) Accounted for travel/lodging costs for some but not all cancer types Did not account for QOL secondary to IQ and hearing loss Little mention of RT doses, was performed prior to dose de-escalation to neuraxis Sensitivity analysis: appropriate variability in costs; variations in estimated risk of adverse effects (IQ loss & GH deficiency) most related to changes in costs of PBT vs. CRT
Mailhot-Vega, USA ³⁴ 2012	Monte Carlo	Linear relationship between IQ reduction (average 10 points) and wage decrease (productivity) No other diseases impact death other than heart disease and second cancer No capital investment, labor, or operational costs included	5 year old children with medulloblastoma	PBT vs. IMRT	\$80,211 (€71,652) PBT \$112,790 (€100,755) IMRT	17.37 PBT 13.91 IMRT	-	PBT superior due to decrease in adverse effects Started to track post-treatment health benefits/costs at age 18 Did not take other endocrine disorders from neuraxial radiation into account No pediatric QOL data Sensitivity analysis: appropriate variability in costs; IMRT with increased costs in nearly all (96%) simulations

Hirano, Japan ³⁶ 2012	Markov	Cochlear RT doses calculated based on institutional data QOL data used for hearing loss Used general death rates to account for other mortality Operational costs included, but no capital investment or labor costs	6 year old children with medulloblastoma	PBT vs. IMRT	-	23.44 PBT 22.46 IMRT	\$11,773 (€10,517)/QALY, \$20,150 (€18,000)/ QALY, or \$21,716 (€19,399)/ QALY depending on quality of life scale used	Specifically assessed cochlear dose-related hearing loss Did not include IQ, productivity/wage loss Operational costs were simply median costs in Japanese facilities regardless of type/area of treatment Sensitivity analysis: relatively high variability; though PBT is appropriately cost-effective, widest variability based on discount rate (3% used in study with range 0-7%)
Mailhot-Vega, USA ³⁵ 2012	Markov	Linear relationship between hypothalamic RT dose and risk of GH deficiency GH costs include that for medication & office visit PBT costs were \$160,000 out-of-pocket more than photons GH deficiency does not impact death whatsoever No capital investment, labor, or operational costs included	4 and 12 year old children with brain tumors requiring hypothalamus radiation dose	PBT vs. IMRT	Various costs depending on age and hypothalamic dose / growth hormone deficiency	Various QALYs depending on age and hypothalamic dose / growth hormone deficiency	Various ICERs depending on age and hypothalamic dose / growth hormone deficiency	PBT more superior if hypothalamic dose difference greater between IMRT and PBT Assumed same costs for growth hormone in adulthood and childhood Did not include IQ or productivity/wage loss Analysis limited to growth hormone deficiency only without accounting for survival outcomes (toxicity only) Sensitivity analysis: appropriate variability in costs; PBT is more cost-effective unless its costs are \$580,000/€546,731 (for 12-year-old) or \$725,000/€683,414 (for 4-year-old) higher than IMRT

Table 5. Summaries of available studies examining cost-effectiveness for pediatric tumors. PBT, proton beam radiotherapy; IMRT, intensity-modulated radiotherapy; QOL, quality of life; CRT, conventional radiotherapy; QALYs, quality-adjusted life-years; ICER, incremental cost effectiveness ratio; GH, growth hormone.

Reference, Country, Year of Cost Analysis	Key Assumptions of Model	Methodology	Cancer Details	Therapy Comparisons	Total Costs	QALYs	ICER	Conclusions & Criticisms
Peeters, Holland ¹⁷ 2007	Assumed facilities of either 2-room photons, 3-room PBT, or PBT/carbon ion (3 rooms) with average lifetime 30 years Investment capital, operational, and labor costs from literature and existing business plans, including interest and replacing linacs every 10 years	Cost estimates from literature	Skull base chordomas	PBT vs. CSRT	€30,530 (\$36,941) PBT, €13,970 (\$16,904) CSRT	-	-	PBT not cost-effective Compared with photon facility, PBT costs increase by 3.2 and particle facility by 4.8 Assumed linear relationship between cost/fraction and number of fractions Lack of clarity on types of tumors treated at certain frequencies (or lack thereof) Not a “cost-effectiveness” analysis, without assessment of outcomes or toxicities at all Sensitivity analysis: performed for operational costs and not cancer specifically; appropriate variability in costs; treatment timings/capacity potentially most variable
Moriarty, USA ⁴⁶ 2011	Post-treatment health states of recurrence, distant metastases, and death from cancer/any cause All metastatic cases die of disease QOL analysis incorporated, but no data available for QOL for recurrence/metastatic disease states No capital investment, labor, or operational costs included	Markov	10,000 patients age 59 with intraocular melanoma	PBT vs. plaque brachytherapy vs. enucleation	PBT \$24,894 (€22,238) Brachytherapy \$28,662 (€25,604) Enucleation \$22,772 (€20,342)	2.938 PBT 2.994 brachytherapy 2.918 enucleation	Compared to enucleation, \$77,500 (€69,230)/QALY brachytherapy & \$106,100 (€94,779)/QALY PBT	Wide variations in results; minimal differences in QALY and little cost differences Effects only analyzed for 5 years post-treatment Assumed that local recurrences were not salvageable and necessarily led to metastases and/or death Sensitivity analysis: high variability with different scenarios presenting each option as most cost-effective in various settings; methodology of sensitivity analysis was only one-way and not multivariate

Table 6. Summaries of available studies examining cost-effectiveness for ocular and skull base tumors. PBT, proton beam radiotherapy; CSRT, conventionally-fractionated stereotactic radiation therapy; QALYs, quality-adjusted life-years; ICER, incremental cost effectiveness ratio; QOL, quality of life.



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